EUROPEAN–TAIWAN CANCER SCREENING CONFERENCE 2026

Screening and Prevention of Liver Cancer

Future Strategies for the European Union from Taiwan's Experience

Vaccinate

Cure

Screen



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THE PROBLEM

The cancer Europe is losing ground on

54,600

liver-cancer deaths, EU-27, 2022

6th

cause of cancer death in the EU

~12%

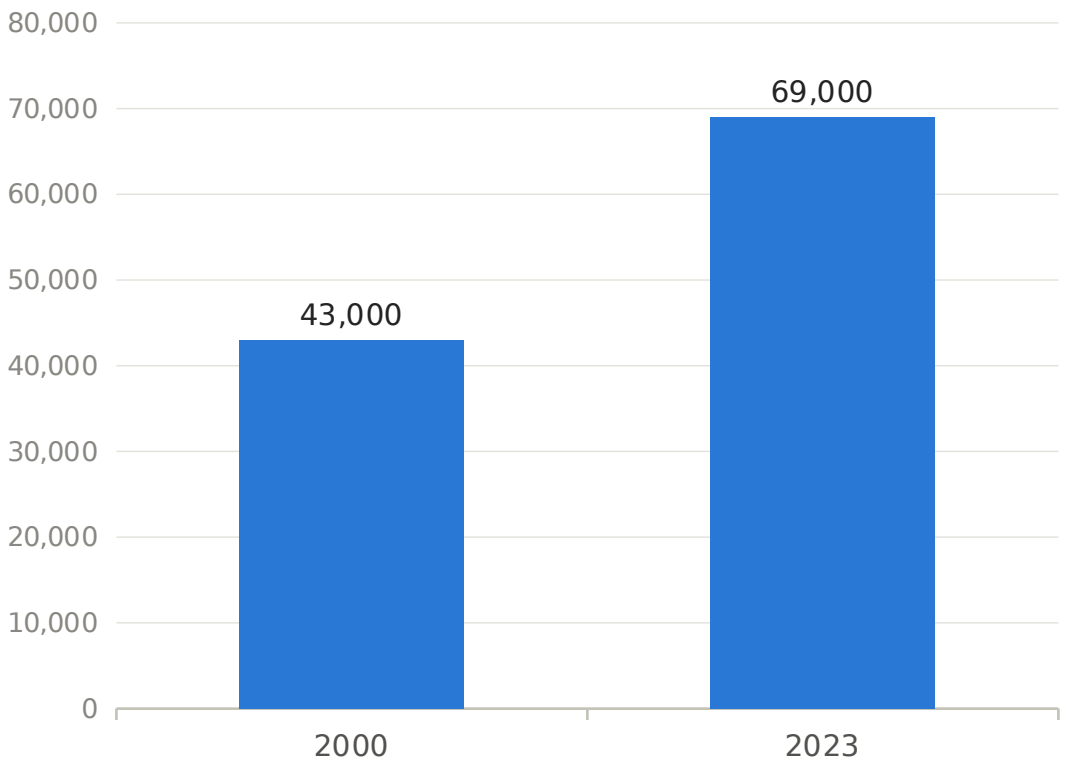
5-year survival in Europe

+22%

further burden growth projected by 2040

While overall EU cancer mortality keeps falling, **liver-cancer deaths are rising.**

Liver-cancer deaths, WHO European region



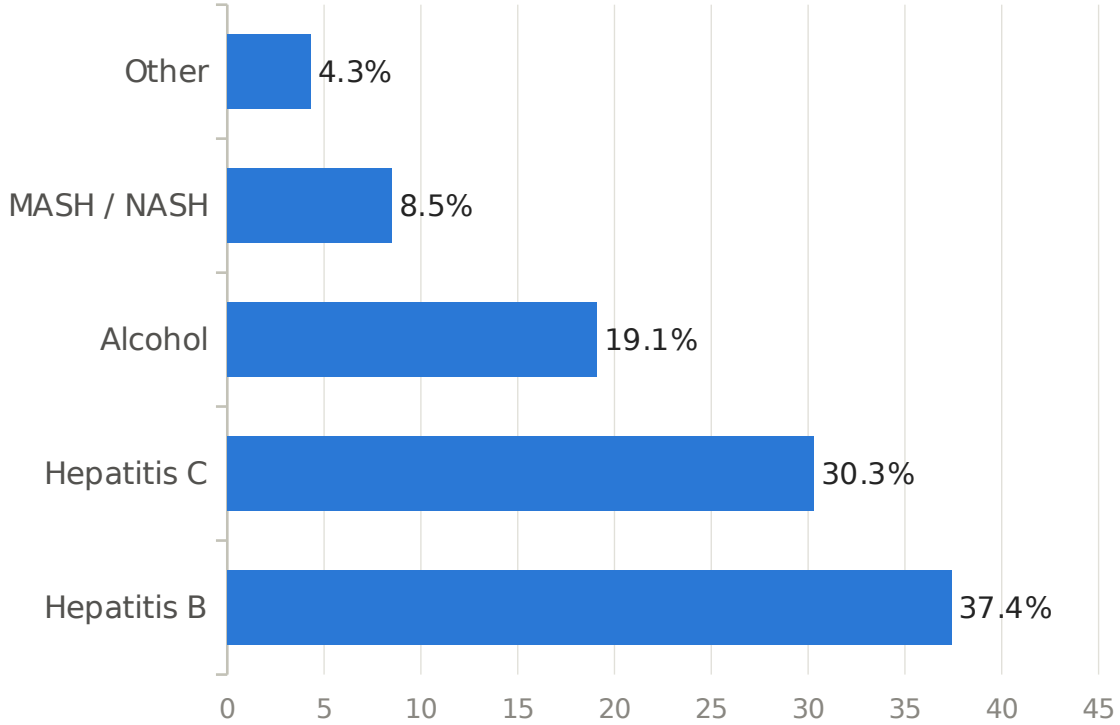
+52% in one generation

Sources: ECIS / European Commission (EU-27, 2022) · 2nd EASL–Lancet Liver Commission, The Lancet 2026 · EUROCARE-5 · UEG Week 2025.

THE PARADOX

The most preventable major cancer — without a programme

Causes of liver-cancer deaths, global (GBD 2021)



Every major cause is identifiable — and actionable — decades before the tumour.

≥ 60% of liver cancers are preventable (Lancet Commission on Liver Cancer, 2025)

- **Causes known:** HBV · HCV · alcohol · MASLD
- **Risk groups identifiable** decades ahead of cancer
- **Proven test:** 6-monthly ultrasound ± AFP
- **No organised EU programme:** liver excluded from the 2022 Council Recommendation on cancer screening

Sources: GBD 2021 · Lancet Commission on Liver Cancer 2025 · EU Council Recommendation on cancer screening, 9 Dec 2022 · EU Group of Chief Scientific Advisors 2022.

TAIWAN'S EXPERIENCE · WHERE IT BEGAN

1980s Taiwan: a hyperendemic baseline

15–20%

of adults were chronic HBV carriers

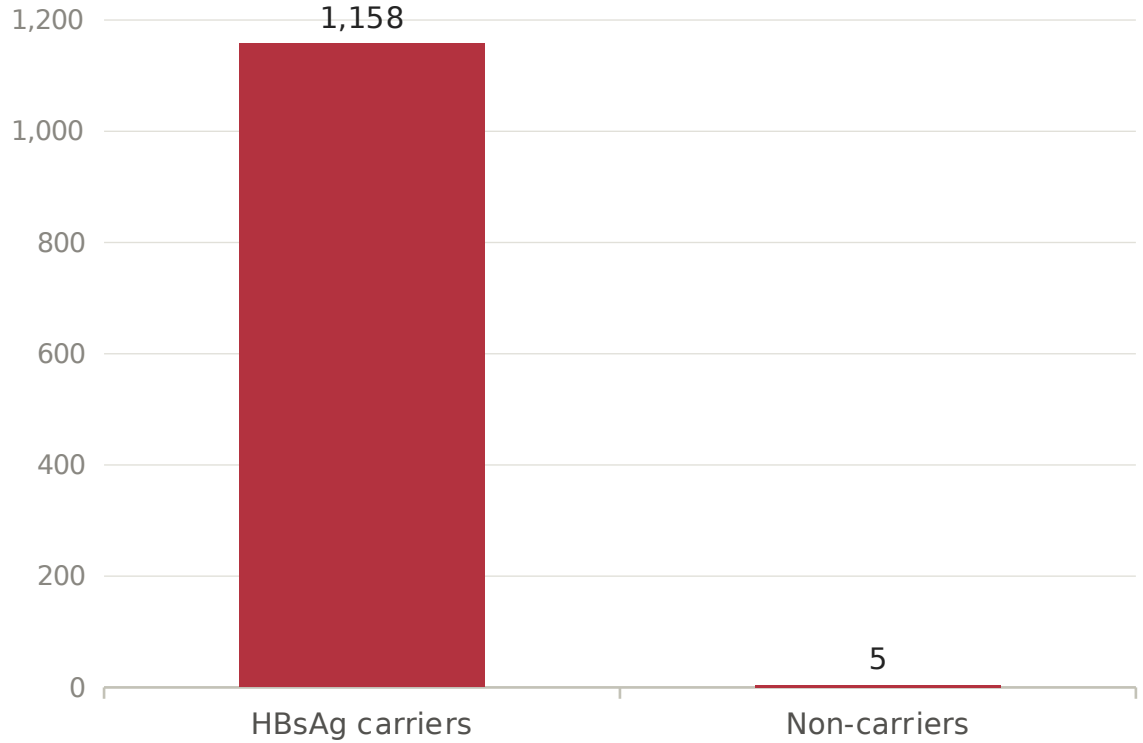
#1

cancer killer in Taiwan for decades

1.8 M

people still live with chronic HBV today — prevention is a marathon

Annual HCC incidence per 100,000 men (Beasley cohort, n = 22,707)



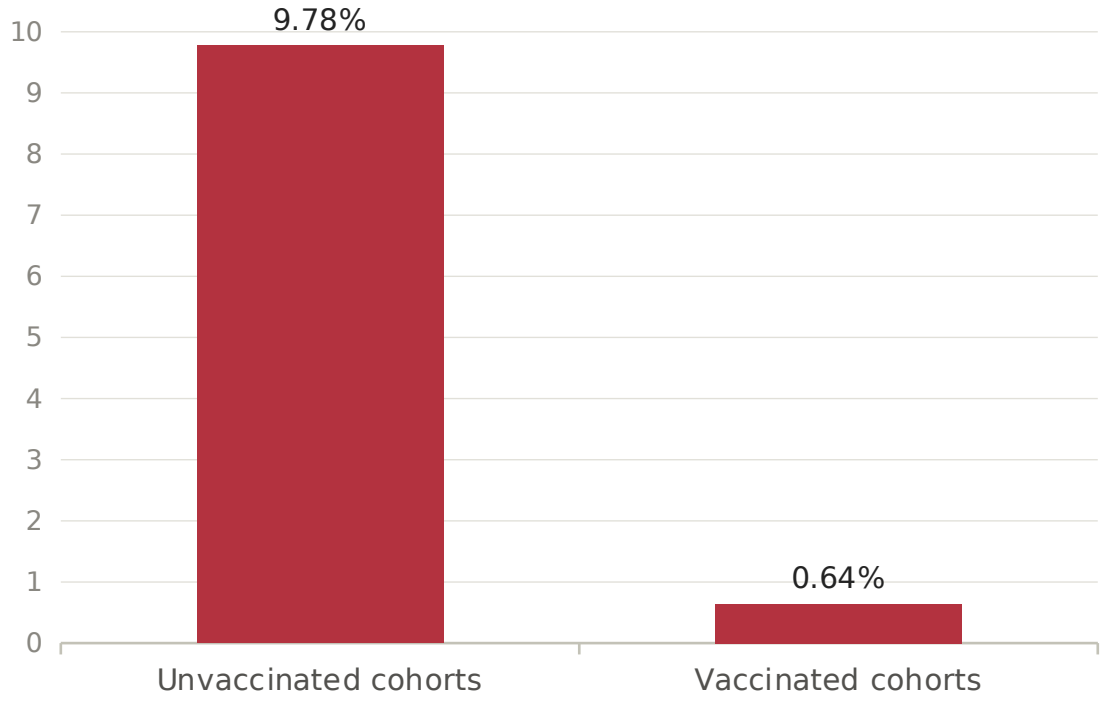
≈ 100× relative risk — the causal proof that made a national policy possible

Sources: Beasley RP et al., Lancet 1981 · Taiwan Business TOPICS / MOHW 2026 · Chen DS, historical seroprevalence surveys.

TAIWAN'S EXPERIENCE · PILLAR 1 — VACCINATE

1984: the world's first universal newborn HBV vaccination

HBsAg carrier prevalence after 35 years of vaccination



national survey, n = 4,717, ages 1–60 (Liver International 2024)

−69%

HCC in vaccinated cohorts (RR 0.31; 1,958 cases, ages 6–29)

−63% / −82%

HCC mortality / chronic liver-disease mortality in immunised birth cohorts

≥ 98%

third-dose coverage sustained since 1989

The first vaccine ever shown to prevent a human cancer.

Sources: Chang MH et al., NEJM 1997; JNCI 2009 · Chang KC et al., Liver Int 2024 · J Formos Med Assoc 2025 · Coalition for Global Hepatitis Elimination 2023.

TAIWAN'S EXPERIENCE · PILLAR 2 — CURE

Hepatitis C eliminated in 2025 — five years ahead of WHO

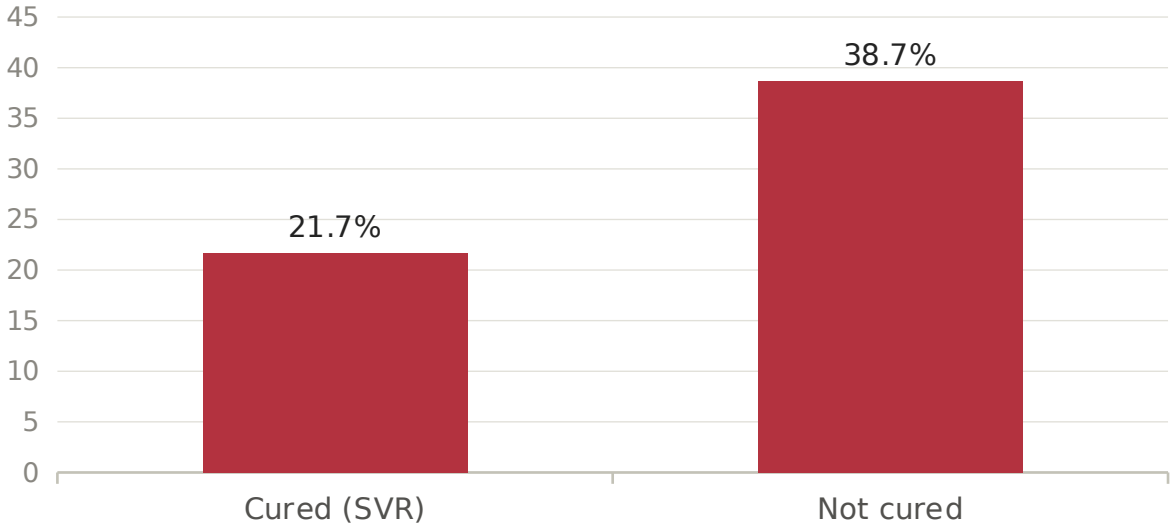
182,000+
treated with state-funded DAAs since 2017

98.3%
real-world cure rate (SVR12, national registry)

90 / 90
diagnosed 90.2% · treated 90.6% — WHO gold tier

€830 M
invested (NT$28.7 bn) — an 'investment in health'

10-year cumulative HCC risk in cirrhosis, by treatment outcome



nationwide cohort, n = 27,577 (Dig Dis Sci 2024)

Micro-elimination first: prisons (100% cure per protocol), dialysis units, PWID, remote villages

Single-payer muscle: one national insurer lifted all fibrosis restrictions by 2019 — treatment free at point of care

Dec 2025: Taiwan announces WHO 2030 goal achieved in 2025; elimination report submitted to WHO WPRO

Sources: Taiwan MOHW, 17 Dec 2025 · TACR registry, Liver Int 2021 · Dig Dis Sci 2024 · National Hepatitis C Program 2016–2025.

TAIWAN'S EXPERIENCE · PILLAR 3 — SCREEN

Find the at-risk, then watch them every six months

Test once, free, for life
national HBV/HCV screening, ages 39–79 — 7.55 M screened

Reflex + registry
automatic HCV-RNA confirmation; every positive enrolled for recall

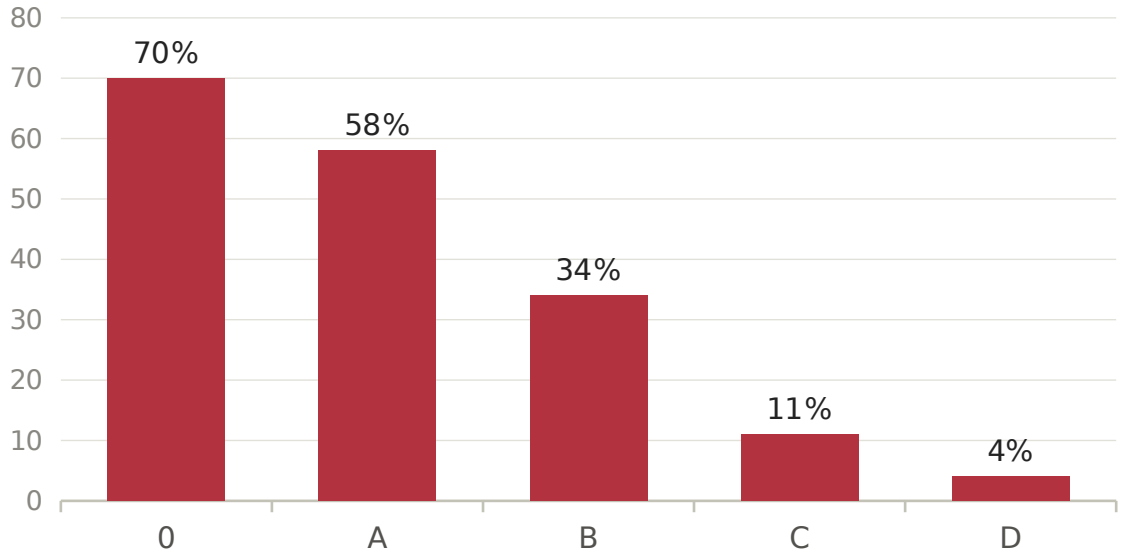
Ultrasound + AFP, every 6 months
for carriers and cirrhosis; 3-monthly if very high risk

Treat early
curative resection / ablation / transplant while it still works

−31% HCC mortality

with invited community ultrasound screening (Changhua, 45–69 y)

5-year survival by stage at diagnosis (BCLC) — why early matters



Taiwan Cancer Registry, 73,817 patients (2011–2019) · earliest-stage (BCLC 0) detection nearly doubled: 6.2% → 11.3%

Sources: Yeh YP et al., Hepatology 2014 · MOHW / HPA 2025 · Viruses 2023 (registry trends) · JAMA Netw Open 2021 (6-monthly interval).

TAIWAN'S EXPERIENCE · OUTCOMES

Four decades of policy bent the curve

1984
universal newborn vaccination

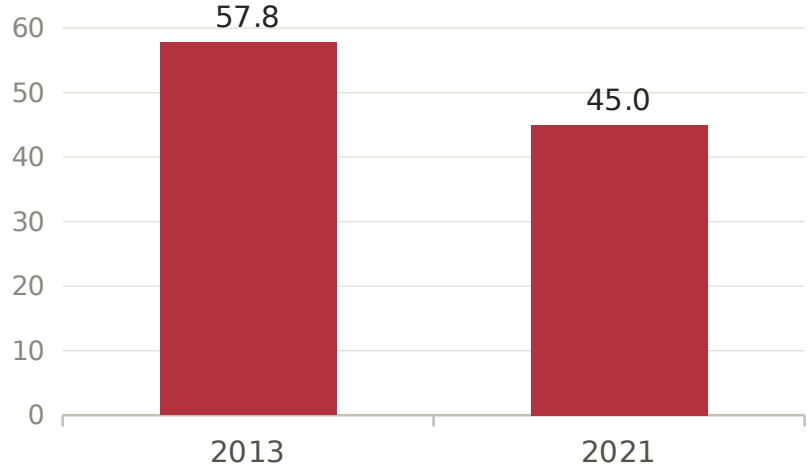
1991
REVEAL cohorts begin

2017
DAAs free for all

2020
adult HBV/HCV screening

2025
HCV eliminated (WHO goal)

HCC incidence per 100,000



14 years
of consecutively falling incidence — Taiwan's only major cancer in decline

7,513
liver-cancer deaths in 2024 — now 2nd, still falling

Out of top 10
chronic liver disease left Taiwan's top-10 causes of death (2023)

−38%
further fall in male incidence projected 2015 → 2035

Sources: HPA Cancer Registry 2023 (rel. Dec 2025) · Clinics & Practice 2024 · MOHW 2024 · J Formos Med Assoc 2019 (projection).

TAIWAN'S EXPERIENCE · LESSONS

What actually made it work

Political commitment, single-payer financing

Screening was never launched without free treatment behind it. The NHI pays; nobody is found positive and then abandoned.

Registries with call & recall

Vaccination, hepatitis, screening and cancer registries are linked. Surveillance is organised, not left to whoever remembers.

Home-grown evidence engines

Beasley, the vaccine trials, REVEAL — national cohorts generated the proof, then policy consumed it within years.

Sequence the attack

Vaccinate the newborn, cure the infected, surveil the at-risk — three pillars, one cascade, each feeding the next.

Synthesis of: J Formos Med Assoc 2025 (national strategy review) · Viruses 2020 (lessons from Taiwan) · NHCP policy papers 2023–2025.

TRANSLATING TO THE EU

Europe's liver cancer is a different disease

Taiwan: one dominant driver

HBV: ~70% of HCC carries hepatitis B — one dominant enemy

Vaccinated: childhood carrier rate 9.78% → 0.64% since 1984

Cured: HCV eliminated 2025 — the virus-first playbook, completed

EU: three drivers at once

HCV legacy: dominant in Southern / Western Europe

Alcohol: up to 50–60% of HCC in Central & Eastern Europe

MASLD: fastest-growing cause — already #1 in Sweden; 35–50% arises without cirrhosis, outside today's surveillance net

Import the system — organised find-treat-surveil — not the virus focus.

Sources: GBD 2021 · J Hepatol 2019 (ARLD-HCC) · Vaz J et al., Int J Cancer 2025 (Sweden) · Hepatology Communications 2025 · AASLD 2024.

TRANSLATING TO THE EU

The gap: opportunistic care, organised nowhere

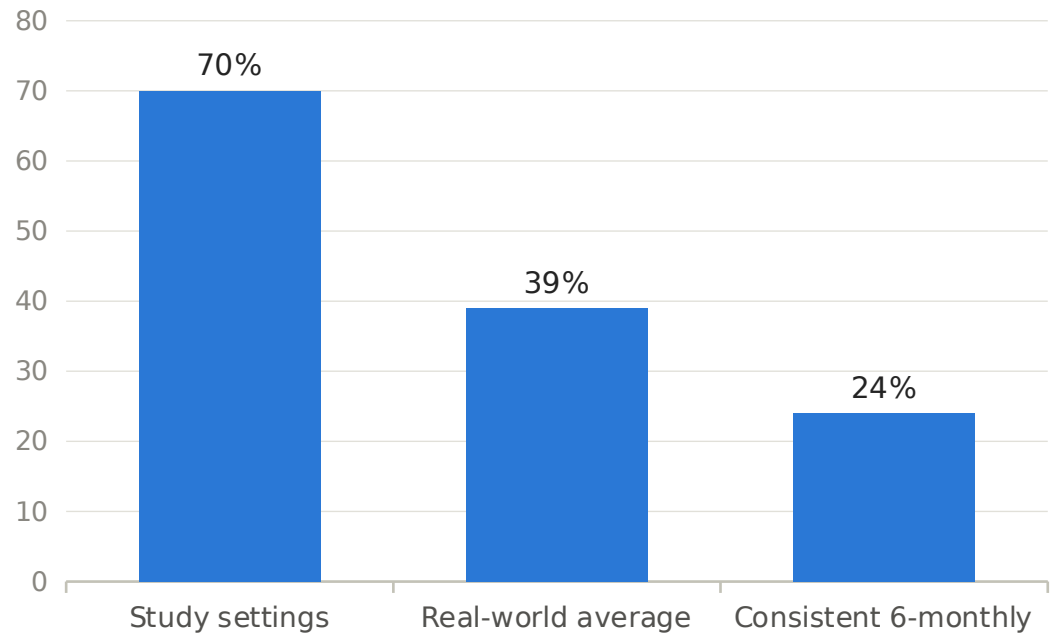
5.4 million in the EU/EEA live with chronic HBV or HCV — ~60,000 deaths every year

Up to 85% / 89% of HBV / HCV infections undiagnosed in some Member States

Off track ECDC 2026: the EU/EEA will miss WHO 2030 hepatitis elimination; only ES · FR · IT · NL on track for HCV

No liver screening programme excluded from the 2022 Council Recommendation — surveillance left to chance

HCC surveillance uptake among eligible patients (%)



60% incurable

of UK cases already beyond cure at first presentation (Lancet Reg Health Eur 2024)

Sources: ECDC 2024/2026 · Polaris Observatory 2025 · J Hepatocell Carcinoma 2024 (adherence meta-analysis) · Lancet Reg Health Eur 2024.

FUTURE STRATEGY 1 · ELIMINATE THE VIRUSES

Find the infected — and treat within weeks, free

1 · Test once in a lifetime

Offer every adult one HBV/HCV test (birth-cohort rollout). Taiwan reached 7.55 M with a free lifetime test.

2 · Reflex + enrol

Reflex RNA confirmation; every positive auto-enrolled in a national registry with recall — no lost letters.

3 · Micro-eliminate

Prisons, PWID, dialysis, migrant communities (EU4Health VH-COMSAVAC shows the model).

4 · Treat free, immediately

DAA / antiviral linkage in weeks. Screening only works when treatment is free — Taiwan's core rule.

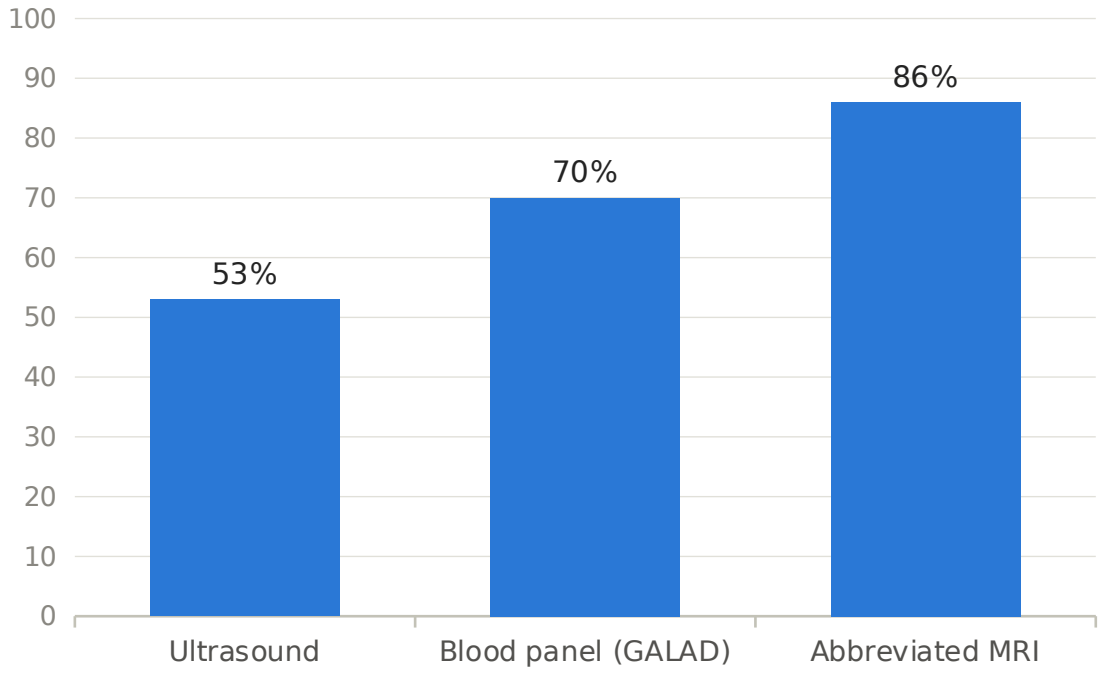
EU target, borrowed from Taipei: **90% diagnosed · 90% treated by 2030** — Taiwan hit both in 2025. The 2024 Council Recommendation on vaccine-preventable cancers (HBV) is the policy hook.

Sources: WHO GHSS 2030 targets · MOHW 2025 · EU4Health VH-COMSAVAC · Council Recommendation on vaccine-preventable cancers, June 2024.

FUTURE STRATEGY 2 · ORGANISED, RISK-BASED SURVEILLANCE

Surveillance smart enough for Europe's aetiology

Early-stage HCC detection sensitivity (published estimates, %)



different cohorts — indicative, not head-to-head (Acad Radiol 2024; Gastroenterology 2024; meta-analyses 2024–25)

Triage by risk score: aMAP / PAGE-B (C-index ≈ 0.82) decide who needs watching — and who safely doesn't

AI-assisted ultrasound: Taiwan is deploying 50 AI ultrasound systems to remote areas in 2026 — quality at scale

LIVERAIM: the EU's own engine — 100,000-participant, 6-country RCT of AI-based early detection

EASL 2025 already endorses risk-based surveillance — what's missing is the organised programme around it: registries, call–recall, QA

Sources: EASL CPG, J Hepatol 2025 · Fan R et al., J Hepatol 2020 (aMAP) · Acad Radiol 2024 (AMRI) · LIVERAIM / IHI 2024 · Taipei Times, Jun 2026.

FUTURE STRATEGY 3 · PREVENT THE NEXT WAVE

The non-viral frontier: alcohol and MASLD

By 2050

1 in 3

liver cancers will stem from alcohol (21%) or MASH (11%)

Lancet Commission on Liver Cancer, 2025

Alcohol policy is cancer policy. Europe drinks more than any other region; 7.1% of EU cancers are alcohol-attributable. Pricing, labelling and availability rules are liver-cancer screening's silent partner.

Find MASLD before cirrhosis. MASLD-HCC is projected to rise 47–130% by 2030, and much of it arises without cirrhosis. LiverScreen (30,000 people, 9 countries) shows FibroScan case-finding works at population scale.

Taiwan is converging on the same problem. Its 2026 national plan adds a non-viral early-detection initiative and a tailored liver-cancer risk model — the playbooks are merging.

Sources: Lancet Commission 2025 · Rumgay H et al., Lancet Oncol 2021 · LiverScreen / Nat Med 2023 · Taipei Times & AmCham TOPICS 2026.

PUTTING IT TOGETHER

A liver-cancer roadmap for the EU, 2026–2040

2026–2028
Prove & prepare

- EU feasibility action on liver screening (Beating Cancer Plan mid-term)
- Birth-cohort HBV/HCV test-and-treat scale-up
- National registries + call–recall build-out

2029–2033
Organise

- Risk-based HCC surveillance programme in every Member State
- Biomarkers / abbreviated MRI as second line (LIVERAIM read-out)
- AI-QA'd ultrasound networks; MASLD case-finding via FibroScan

2034–2040
Deliver

- Hepatitis eliminated as a public-health threat
- >50% of HCC detected at a curable stage
- Liver-cancer mortality curve bent — Europe's version of Taiwan's 14-year decline

Measure like Taiwan: 90/90 hepatitis diagnosed & treated · >50% early-stage detection · mortality bent by 2035

Framing: Europe's Beating Cancer Plan 2021 · 2nd EASL–Lancet Commission 2026 · WHO GHSS 2022–2030 · Taiwan MOHW milestones as benchmarks.

CONCLUSIONS

Liver cancer is a policy-sensitive disease

Taiwan proved it. Vaccinate, cure, screen — HCV eliminated 5 years early, and 14 consecutive years of falling liver-cancer incidence.

Europe has the tools — not the programme. Risk scores, biomarkers, abbreviated MRI, AI ultrasound, LIVERAIM. What's missing is organisation: registries, call–recall, guaranteed treatment.

The window is open. EASL–Lancet Commission 2026, the Lancet HCC Commission, ECDC's warning and the Beating Cancer Plan mid-term all point one way: put liver on Europe's screening agenda now.

“Taiwan wrote the proof of concept. The European Union can write the sequel.”

Thank you · 謝謝 · Merci · Danke

FOR THE PROCEEDINGS

Key references

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